Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 · Energy storage devices, hydrogen energy, and sensors · Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II))

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II))
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II))

1 Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II))

Study focus: Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000026 · EV0000027

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module five on energy storage devices.

Key details

- We are talking about batteries.
- We discussed two battery characteristics that is voltage and current.
- And some of them which we are going to study today are capacity, energy storage density, cycle life, energy efficiency, shelf life, tolerance to service conditions, power density and energy density.
- What is the definition?
- The charge or amount of electricity that can be obtained from the battery is known as capacity of the battery, and the unit in which it is expressed is ampere hours.
- W is the mass of the electroactive material.
- M is molar mass of the electroactive material.
- And n is the number of electrons being exchanged in the reaction.
- 2 plus oxidation should happen at dianode.
- N is going to be 2, and F is the Faraday that is 96500 coulombs per mole.
- We can use to determine the capacity of a battery if we know the amount of the electroactive material present at the electrode.
- How can we say that?
- Because W is figuring in this expression.

- And what is W?
- Just now I told you W is the amount of the electroactive material.
- Bigger is going to be the battery.
- Therefore, we can say that the size of the battery determines the capacity of the battery.
- Now, another condition on which the capacity of the battery depends is the discharge of the battery.
- What does it mean?
- How much current is being drawn from the battery.
- It is not very different from what we have studied before.
- Because capacity is a charge contained in the battery.
- And current multiplied by time is also charge.
- Wnf by M.
- If we know how long the battery lasts when we are drawing a certain amount of current, then we can use this expression to determine capacity of the battery.
- We are seeing how the voltage varies with time for a particular battery.

Example or application from the lecture

- Let us first learn about capacity of a battery.
- The capacity is also given by a current into time.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 05:00–09:59

almost a flat portion and after that it comes to a pos part the voltage is now too small for the battery to.

Key details

- Perform the required function.
- Initially, there will be a drop in the voltage, and then we see that it gives a steady voltage for a long time until it reaches a point.
- After which we declare that battery as dead, if it's a rechargeable battery, we need to recharge it.
- So this is the cutoff voltage because beyond this, every device requires a certain voltage.
- Device is not going to function using this battery, so either it needs to be replaced or it needs to be recharged.
- So we can say that the capacity of the battery, because what we are looking at when we are drawing a certain amount of current, how long the battery will last.
- portion of this curve can give us an indication as to what is the capacity of the battery.
- So if this flat portion is very long, we say that the battery has a good capacity because at this particular current it is giving you it lasts for a longer time.
- If this portion is the flat portion is less then we say that the capacity is not so good.
- Energy storage density is defined as the amount of charge or electricity per unit weight which the battery can hold.

- It is telling the amount of charge.
- Which the battery can hold, but it is now it is become density.
- We are calling it density, so that is per unit weight which the battery can hold.
- So that means if we want to get a good electricity storage density, we have to use the right material for all the components.
- We say that the designing of the battery is very good.
- Electroactive material that is taking part in the reaction at the cathode or the anode.
- But here when we say weight, we refer to weight of the battery on the whole, meaning all the components of the battery have to be taken into account.
- Why lithium is such an important anodic material?
- Why is it such a popular anodic material?
- Because we know it's a lightweight metal.
- Just seven grams of lithium can give you one faraday of charge.
- Whereas if we use lead, we know that lead is also a very popular anonic material.
- like lead acid batteries etc which are which are used quite a lot so but here we see that we require 104 grams of lead to produce the same amount of charge that is one farity so therefore lithium has become a very popular aerodic material.
- Now coming to the next battery characteristic, cycle life.
- Such batteries are referred to as secondary batteries.
- These secondary batteries can be again and again charged, but they also have a lifetime.
- The electroactive when it is when the battery is discharging, the electroactive material is consumed.
- But when the electroactive material is completely consumed, we can again.

Example or application from the lecture

- Now, let us come to the next battery characteristic that is energy storage density.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 10:00–14:59

recharge the battery and when we recharge the battery this electroactive material is getting regenerated.

Key details

- That means the cycle can go on and on because you can charge and discharge, charge and discharge as many times as we want, but we know that this is not true.
- Every secondary battery also has a life.
- How will we define it?
- The number of charge-dischar cycles that are possible before the battery fails.
- How many times can we charge and discharge the battery?
- That 500 times the battery can be charged and discharged.
- The battery can go through before it fails.
- Why do secondary batteries have a limited cycle life?

- Theoretically, you can charge and discharge, charge and discharge, but this does not go on forever.
- We know that there is a limited life for a particular battery, even though it is a secondary battery.
- For this is corrosion of the contact points.
- We know that we are using as electrolytes in the form of acids, alkali, salt solution, all those are very corrosive materials, and therefore the contains with time could get corroded.
- Reason why we see that the batteries have a limited life.
- Going to come out and electrons will be generated and the electroactive species is getting consumed.
- has to come back and form a nice layer around the around the current collector.
- Initially, yes, there will be nice layers of this electroactive material, and you will get something like in this form.
- But as time progresses, you go on using the battery for a long time.
- We find that this material instead of being on the current collector it will start to shut down.
- This electroactive material during the charging process has not gone back to the current collector and therefore what will happen the this material which has shut down is not available for the next discharging cycle.
- The electroactive material is.
- Able to undergo reaction as long as it is around the current collector.
- It is available for the discharging.
- But once it is not on the current collector and a shutdown like this, the electroactive material slowly goes on decreasing, and we find that the battery will become weaker and weaker.
- So first we talked about corrosion, next.
- What do we mean by this?

Example or application from the lecture

- Let us draw a battery here and see what exactly happens inside the battery.
- Suppose the battery is discharging, all this is the electroactive species present around it, and this is the current collector.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 15:00–19:59

When the battery is discharging, we find this electroactive material gets away from the current collector.

Key details

- When you are charging, again as I said in the previous point, this material has to come back to the current collector.
- Forming nice layers around the current connector, we find that with time these dendrite like growths start.

- the battery shows tree like growth or they are called dendrite like growth.
- Anode, the same thing can happen at the cathode also.
- Even there is a separator in between, but they may pierce through the separator and what we find is what will happen when cathode and another come in contact with each other, there is going to be short circuiting and the battery is going to fail.
- The electroactive material did not get back onto the current collector.
- Instead, we find dendrite like growth.
- Or you can simply say tree like structures.
- Have to maintain low inter electrode distance.
- Coming to the next battery characteristic, energy efficiency of a battery.
- Percentage of energy efficiency would be energy released on discharge by the energy required for charging multiplied by 100.
- Charging this is again applicable to secondary batteries.
- Or this definition of energy efficiency is applicable to secondary battery.
- It is never 100%.
- And this efficiency depends on the efficiency of the electrode reactions.
- If there is any hampering in the electrode reactions, or they are going slow, or there is some kind.
- Problem the anode efficiency, the cathode efficiency is not good, then we find that the energy efficiency also falls down.
- Coming to the next characteristic that is shelf life.
- Many times we keep batteries at home, and especially the dry cell, we find that when we want to use the battery, it is not working anymore.
- That what is it, what is responsible for the battery being dead when it is not at all being when it has not even been used.
- That characteristic is called shelf life.
- Shelf life, how will we define shelf life?
- The maximum time, the maximum time for which you can store the battery, and yet it will give you the specific performance.
- How long can you keep the battery without using it?
- And yet it gives you the specific performance which is required.
- But many times we find that batteries are dead by the time we want to use, and of course, this is we are talking of a primary battery like dry cell, we find that.
- They are not useful anymore and they cannot be used anymore and have to be discarded.
- What could be the reason for this?
- The reason is there is something called as self-discharge, which happens inside the battery.
- The moment the components are assembled.
- the reaction in the battery starts.
- So therefore, or there could be other secondary reactions also happening which are going to consume the electroactive material and therefore when we leave it for a long time without using it then the battery all the material could have been used up or there could also be corrosion and leaking of the battery.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 20:00–24:59

the shelf for a battery having shelf life.

Key details

- Of course it is very important for a primary battery to have a good shelf life.
- Why because so that even if it is stored for some time we can still use it as primary batteries are one time use batteries.
- For secondary batteries also have a shelf life, but the good thing about secondary batteries is they can be recharged.
- Problems like corrosion and all will of destroy the battery.
- Now coming to the next point, tolerance to service conditions.
- The battery has to be tolerant to different service conditions.
- Be available, I mean it should operate at very low temperatures or at the high temperature at which the.
- The motor gets heated up, I mean the motor is going to heat up the machine, and therefore it has to tolerate that high temperature.
- or shock because when the when you're moving the car when the car is moving on the road or in any terrain there will be lot of vibration and shocks which the battery will also have to face.
- So therefore the battery should be designed such a way that it should be tolerant.
- To whatever service conditions you are going to use it for, whatever application it is going to be used for, the service conditions, the battery should be able to tolerate those service conditions.
- Now energy density is defined as the amount of energy which is available from the battery divided by mass of the battery, or you can say weight of the battery.
- Again, when we.
- We say density, we should remember the weight includes weight of all the components of the battery.
- Divided by weight of the battery.
- And the what is the current the battery is giving us multiplied by the voltage at which it is operating, multiplied by the time for which the battery will.
- Watt hours per kg, or we can say kilowatt hours per kg.
- Depending on what is the application, the unit can be varied, but nevertheless, it is power into time divided by weight of the battery.
- Coming to the last battery characteristic that is power density.
- Pow density is defined as the power available from the battery divided by mass of the battery or weight of the battery.
- Again since we are talking of density.
- It should be the weight of all the components of the battery or mass of all the components of the battery.
- Power is always current into voltage.
- being you have used there multiplied by voltage of.
- The battery divided by weight of the battery, mass of the battery.

- That would be power density of the battery and the unit will be watts per kg.
- I said for energy density, it can also be kilowatt per kg, watts per gram, depending on the application and depending on the size of the battery, and depending on what power it is generating.
- We can choose the correct unit, but the unit is going to be that of power divided by.

Example or application from the lecture

- For example, if there is a variation in temperature.
- The temperature range has to be very large for, for example, the car batteries, and apart from that, they should also be tolerant to say vibration.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 25:00–29:59

weight of the battery.

Key details

- Both look very similar.
- But power density, as we know, energy is the work done.
- Energy is always work done, and power is at what rate the energy is being produced.
- When the car is going uphill, you need a high power density.
- And when would you require a good energy density?
- How would whether the energy density is good?
- it lasts for a long time before it requires charging.
- Power density will be when it the when the say it is in a car and how long so energy density is how long the car would go before it requires charging and power density.
- How steep a hill can it climb?
- That would be determined by its power density.
- Nevertheless, we have to know.
- See what is the application.
- Some applications could require a very high power density, but some applications could require a low power density.
- Also, energy density or power density, the requirement could be that we need a certain amount of energy density for a long period of time or.
- a high energy density for a very short period of time.
- Whereas if you're running the car, if you're running an electric vehicle, then you would require a high energy.
- Density, a constant energy density for a long time.
- density and all these battery characteristics are very important and it's not necessary that all these characteristics should be at the high value for every for an application.
- Sometimes we require.
- low power density is fine.

- But then there are other requirements like it should be able to tolerate the body temperature, then it should have it should be very accurate voltage.
- And it should be very, very reliable.
- It should last for a long time.
- So in the next class.
- We are going to look at some modern batteries and then we will see how all these battery characteristics have been used for these.
- or what are these battery characteristics or that particular battery.
- So in the next class we will talk about zinc air battery.

Example or application from the lecture

- When would you require a high power density when you when you say for example when you're when the car is going uphill.
- For example, in the flash of a camera, we need very high energy density, but for a very short period of time.
- For example, in a pacemaker, we require we don't require very high power, we require just few micro watts.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00007	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00009	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

C Revision and retrieval practice